

Application of AI in Natural Language Processing (NLP)

Introduction

- It's a branch of Artificial Intelligence, aimed to make machines read, understand and derive meaning from human languages.
- Giving computers the ability to understand text and spoken words in much the same way human beings can.
- It helps machines process and understand the human language so that they can automatically perform machine translation, summarization, document classification, and spell check.
- It combines computational linguistics and rule-based modeling of human language with statistical, machine learning, and deep learning models.
- These technologies enable computers to process human language in the form of text or voice data.

Components of NLP

- **Morphology**- is concerned with the interplay between words and their relationship with other words. We can say It is a study of construction of words from primitive meaningful units
- **Syntax** – It refers to arranging words to make a sentence. It also involves determining the structural role of words in the sentence and in phrases.
- **Semantics** – It is concerned with the meaning of words and how to combine words into meaningful phrases and sentences.
- **Pragmatics** – It deals with using and understanding sentences in different situations and how the interpretation of the sentence is affected.
- **Discourse** – It deals with how the immediately preceding sentence can affect the interpretation of the next sentence.
- **World Knowledge** – It includes the general knowledge about the world.

NLP pre-processing tasks

- **Tokenization:** breaks down text into smaller semantic units or single clauses
- **Bag of Words:** can represent a sentence as a bag of words vector (a string of numbers).
- **Part-of-speech-tagging:** marking up words as nouns, verbs, adjectives, adverbs, pronouns, etc.
- **Stemming and lemmatization:** standardizing words by reducing them to their root forms
- **Stop word removal:** filtering out common words that add little or no unique information, for example, prepositions and articles (at, to, a, the).

Tokenization

- Tokens are the building blocks of Natural Language. Processing the raw text happens at this level.
- Its a way of separating a piece of text into smaller units called tokens (words, characters, or subwords) .
- Task of cutting a text into pieces called tokens, and at the same time removing certain characters, such as punctuation.
- The most common way of forming tokens is based on space.
- Tokens are used to prepare a vocabulary. Vocabulary refers to the set of unique tokens in the corpus.

Types of Tokenization

- **Word Tokenization**

- It splits a piece of text into individual words based on a certain delimiter.
- For example, consider the sentence: **"I love India"**.
- Assuming space as a delimiter, the tokenization of the sentence results in 3 tokens –

I love India

- **Character Tokenization**

- Splits a piece of text into a set of characters
- Eg. **Shivaji** → **S-h-i-v-a-j-i**

- **Subword Tokenization (n-gram characters)**

- Splits the piece of text into subwords
- Eg. **Greatest** can be segmented as **Great-est**.

Limitations

- Splitting on blank spaces may break up what should be considered as one token, example: New Delhi or West Bengal
- **Tokenization can remove punctuations**
- In the case of periods that follow abbreviation (e.g. Dr.), the period following that abbreviation should be considered as part of the same token and not be removed.
- The tokenization process can be particularly problematic when dealing with biomedical text domains which contain lots of hyphens, parentheses, and other punctuation marks.

Bag of Words

- A bag of words is a representation of text that describes the occurrence of words within a document
- Occurrence matrix for the sentence or document. Keeps track of word counts and disregard the grammatical details and the word order.
- The model is only concerned with whether known words occur in the document, not where in the document.
- These word frequencies or occurrences are used as features for training a classifier.

Eg. S_1 : I love programming

S_2 : Python is my favorite programming language

Sentence	I	love	programming	Python	is	my	favorite	language
S_1	1	1	1	0	0	0	0	0
S_2	0	0	1	1	1	1	1	1

Drawbacks

- If the new sentences contain new words, then vocabulary size would increase and thereby, the length of the vectors would increase too.
- Additionally, the vectors would also contain many 0s, thereby resulting in a sparse matrix
- All words have the same importance. No semantic information preserved.
- No information on the grammar of the sentences nor on the ordering of the words in the text is retained
- Stop words (like “the” or “a”) add noise to the analysis and some words are not weighted accordingly (“Newton” weights less than the word “is”).
- “Today is a beautiful day”

Term Frequency - Inverse Document Frequency (TF-IDF)

- TF-IDF is a statistical measure that evaluates how relevant a word is to a document in a collection of documents.
- This is done by multiplying two metrics: how many times a word appears in a document, and the inverse document frequency of the word across a set of documents.
- It works by increasing proportionally to the number of times a word appears in a document, but is offset by the number of documents that contain the word.
- Words that are common in every document, *this*, *what*, and *if*, rank low, since they don't mean much to that document in particular.
- Some semantic information is preserved as uncommon words are given more importance than common words.

TFIDF...

"It's looking good "

- Here **good** will have importance

"It's looking bad "

- Here everything is same except instead of good we are using bad , which completely changes the meaning of the sentence.
- It evaluates the words used in a sentence and measures it's usage compared to words used in the entire document. it's a score to highlight each word's relevance in the entire document. Calculated as:

IDF = $\text{Log}[(\# \text{ Number of documents}) / (\text{Number of documents containing the word})]$

TF = $(\text{Number of repetitions of word in a document}) / (\# \text{ of words in a document})$

- **TF** answers questions like — *how many times is good used in that entire document*, and **IDF** answers questions like *how important is the word good in the entire list of documents, is it a common theme in all the documents*.

Part-of-speech (POS) tagging

- **POS tagging** is the process of marking up a word in a corpus to a corresponding part of a speech tag, based on its context and definition.
- A particular word may have a different part of speech based on the context in which the word is used.
Example: In the sentence “**Give me your answer**”, *answer* is a Noun, but in the sentence “**Answer the question**”, *answer* is a verb.
- To understand the meaning of any sentence or to extract relationships and build a knowledge graph, POS Tagging is a very important step.

Example

Input: **Everyone back to work.**

- Everyone --- Pronoun
- Back --- adverb
- to --- preposition
- Work --- verb

Output: [('**Everyone**', NNPS) , ('**back**', RB), ('**to**', TO), ('**work**', VB)].

POS Tagging Techniques

- **Lexical Based Methods** — Assigns the POS tag the most frequently occurring with a word in the training corpus.
- **Rule-Based Methods** — Assigns POS tags based on rules. For example, we can have a rule that says, words ending with “ed” or “ing” must be assigned to a verb. Rule-Based Techniques can be used along with Lexical Based approaches to allow POS Tagging of words that are not present in the training corpus but are there in the testing data.
- **Probabilistic Methods** — This method assigns the POS tags based on the probability of a particular tag sequence occurring. Conditional Random Fields (CRFs) and Hidden Markov Models (HMMs) are probabilistic approaches to assign a POS Tag.
- **Deep Learning Methods** — Recurrent Neural Networks can also be used for POS tagging.

Stemming

“The process of reducing inflection towards their root forms are called Stemming, this occurs in such a way that depicting a group of relatable words under the same stem, even if the root has no appropriate meaning”.

- It is a technique in which a set of words in a sentence are converted into a sequence to shorten its lookup. In this method, the words having the same meaning but have some variations according to the context or sentence are normalized.
- In another word, there is one root word, but there are many variations of the same words. For example, the root word is “eat” and its variations are “eats, eating, eaten and so on”.

For example

He was riding.

He was taking the ride.

- Stemming is used to categorize the same type of data by getting its root word.
- Refers to the process of slicing the end or the beginning of words with the intention of removing affixes. Affixes that are attached at the beginning of the word are called prefixes (e.g. “Bio” in the word “Biology”) and the ones attached at the end of the word are called suffixes (e.g. “ful” in the word “wonderful”).

Advantages & Limitations

- Shortening the vocabulary space, thus drastically improving the size of the index (or feature space).
- Stemming is used in information retrieval systems like search engines.
- Not all the words can be stemmed to their right root word.
- It can't relate words which have different forms based on grammatical constructs, like *is*, *am*, *be*, all represent the same root verb, *be*. But stemming can't prune them to the common form. The word *better* should be resolved to *good*, but stemmers would fail to do that.

Lemmatization

- Lemmatization technique is like stemming. The output after lemmatization is called 'lemma', which is a root word rather than root stem.
- Process of finding the lemma of a word depending on its meaning and context.
- Objective is to reduce a word to its base form and grouping together different forms of the same word.

Example, verbs in past tense are changed into present (e.g. "went" is changed to "go") and synonyms are unified (e.g. "best" is changed to "good"), hence standardizing words with similar meaning to their root.

"Lemmatization resolves words to their dictionary form (known as lemma) for which it requires detailed dictionaries in which the algorithm can look into and link words to their corresponding lemmas".

Example

- The words “running”, “runs” and “ran” are all forms of the word “run”, so “run” is the lemma of all the previous words.

Caring → Care

Lemmatization

Caring → Car

Stemming

- Lemmatization also takes into consideration the context of the word in order to solve other problems like disambiguation, which means it can discriminate between identical words that have different meanings depending on the specific context.
- By providing a part-of-speech parameter to a word (whether it is a noun, a verb, and so on) it's possible to define a role for that word in the sentence and remove disambiguation.

Stemming and Lemmatization

- Text normalization techniques for Natural Language Processing.
- Stemming is faster as it cuts words without knowing the context, while lemmatization is slower as it knows the context of words before processing.
- Stemming just removes or stems the last few characters of a word, often leading to incorrect meanings and spelling. Lemmatization considers the context and converts the word to its meaningful base form, which is called Lemma. Sometimes, the same word can have multiple different Lemmas
- Lemmatization is a much more resource-intensive task than performing a stemming process.
- It requires more knowledge about the language structure than a stemming approach, it demands more computational power than setting up or adapting a stemming algorithm.

Stop Words

- Stop Words are words which do not contain important significance to the corresponding text.
- Includes common language articles, pronouns and prepositions such as “and”, “the”, “to”, “are” etc. in English.
- Common words that appear to provide little or no value to the NLP objective are filtered and excluded from the text to be processed, removing widespread and frequent terms that are not informative about the corresponding text.
- Stop words can be safely ignored by carrying out a lookup in a pre-defined list of keywords, freeing up database space and improving processing time

Advantages of NLP

- Helps computers to communicate with humans in their languages.
- It is very time efficient.
- Most of the companies use NLP to improve the efficiency of documentation processes, accuracy of documentation, and identify the information from large databases.
- Users can ask questions about any subject and get a direct response within seconds.
- The accuracy of the answers increases with the amount of relevant information provided in the question.
- Allows to perform more language-based data as compared to a human being , without fatigue and in an unbiased and consistent way.

Disadvantages

- May not show context.
- Is unpredictable
- Unable to adapt to the new domain, and it has a limited function, built for a single and specific task only.
- Complex Query Language- the system may not be able to provide the correct answer if the question is poorly worded or ambiguous.
- Doesn't have a user interface which lacks features that allow users to further interact with the system